

PHLOTON

Reliable cold chain for pharma, labs, blood banks and clinics: active, off-grid, connected, with a regulatory path a buyer can audit.

THE DEVICE, IN THREE WORDS

Active. Off-grid. Connected.

A solar-charged, connected carrier that holds vaccines, blood and biologics at 2–8 °C where the grid ends. Fifty units in service in India; built for the world.

Prerit Mittal, founder • prerit@phlton.com • phlton.com



2 L units in assembly, Bengaluru

Every 2–8 °C product now has a last mile to lose it on.

38% of pharma products need cold chain, up from 26% in 2017. Biologics are more than half the clinical pipeline, and a single vial can cost about \$760 — the price of the device that carries it. Between depot and patient, that value rides in an ice box that records nothing.

38%

of pharma products now need cold chain — 26% in 2017

IQVIA, 2025

\$760

for one vial of a marketed retinal biologic — about the price of the device

Indian list price

7–25%

of vaccine value lost to temperature failures alone

IQVIA, 2025

WHO HAS ALREADY SWITCHED

Beckman Coulter ran diagnostic samples on 7–8 hour routes for six months, reported no excursions and no sample loss, then bought. **Roche India** is starting depot-to-hospital cold-chain-as-a-service in Kolkata. **Raktamitra** blood bank moved blood bags with no haemolysis.

WHAT THE BUYER GETS

A \$700 device with a five-year warranty, \$60 a year of telemetry and per-trip compliance reports, and a path procurement can read: CDSCO test licence in hand, **ISO 13485** and CDSCO MD-5 next, CE and WHO PQS E003 funded by this round.

Half of what the world makes never protects anyone.

By WHO's upper estimate, up to half of all vaccines are wasted every year; IQVIA puts the loss to temperature alone at 7–25%. About a billion people in lower-income countries depend on health facilities without reliable power. A cold chain that needs a grid and a freezer cannot follow the medicine to the patient.

50%

of vaccines wasted every year at WHO's upper estimate; up to a quarter lost to temperature alone

WHO, via UNEP 2020 • IQVIA 2025

\$35^B

lost by biopharma yearly to temperature failures

IQVIA Institute estimate, 2019

1^B

people in lower-income countries served by health facilities without reliable power

WHO • World Bank • IRENA, 2023

HOW THE LAST MILE IS SOLVED TODAY

Ice-pack and phase-change-material (PCM) carriers, conditioned in a freezer upstream — or a mains-charged active carrier at about \$2,000.

WHY THAT FALLS SHORT

A passive carrier needs a working freezer, loses about half of what is left each time it is opened (our field rule of thumb), can freeze the payload, and records nothing. A mains-charged active carrier has to come back to a socket to recharge; the grid sets its range.

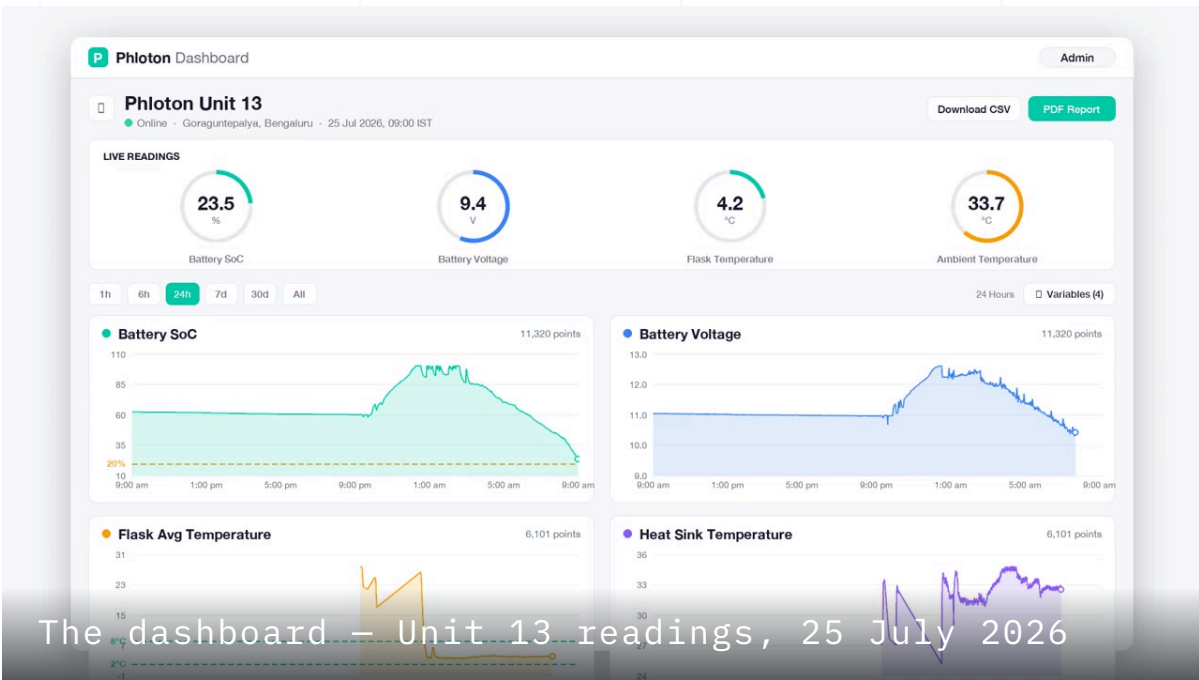
2–8 °C. Designed for sixteen hours at 43 °C. Charged by the sun.

A 3.75 kg, 2-litre device with a solid-state cooling engine: no ice, no freezer, no refrigerant. It holds 2–8 °C, charges from the sun, a vehicle or the wall, and reports every reading over 4G. From a warm start an empty chamber is inside 2–8 °C in fifteen to twenty minutes; the status light turns green when the chamber is in band.

AMBIENT • DESIGN	PAYLOAD • TARGET	HOLD • DESIGN	AUDIT TRAIL
43 °C	3 °C	16 h	Every trip

WHAT THE CUSTOMER SEES

Sixteen sensing points reported over 4G, with an offline buffer for the stretches without signal: each unit's readings and temperature trace, alerts when a reading drifts, the fleet on a map, and a compliance report at the end of every trip.



One engine. Every payload that must stay cold.

Every proven row below runs on the same solid-state cooling engine and controller. A new payload is a new customer, not a new engine; the larger sizes add a coolant loop.

PAYLOAD	HOLD BAND	WHERE IT STANDS
PROVEN IN THE FIELD		
Vaccines	2–8 °C	SELCO Foundation outreach clinics · PCM jackets for open-vial sessions
Blood bags	2–6 °C · never freezes	Raktamitra blood bank, Government Institute of Medical Sciences (GIMS), Noida — transported with no haemolysis (no red-cell damage)
Diagnostic samples	2–8 °C	Beckman Coulter — six months of 7–8 h runs, then a purchase
Specialty biologics	2–8 °C	Roche India, Kolkata — cold-chain-as-a-service, depot to hospital · starting
Veterinary & dairy	2–8 °C	Akshayakalpa · URMUL
NEXT — SAME ENGINE, NEW SET POINT OR FORM FACTOR		
Insulin & antivenom	2–8 °C	2.5 L for primary health centres
Reagents, enzymes, trial kits	programmable	5 L for labs and clinical-trial logistics
Organs & tissue	4–10 °C	10 L · exploratory, with a transplant partner

Active, off-grid, connected. Only one does all three.

The last mile needs three things: active cooling, power without a grid, and a live record. The table scores each option against all three.

	ICE - PACK CARRIER	PCM CARRIER	BLACKFROG EMVOLIO	PHLOTON
Price	~\$75 prequalified + a freezer	varies + a freezer	~\$2,000	\$700 → \$500 at scale
Holdover	set by the ice packs	fixed window, halves with each opening	12 h, verified at 43 °C	16 h at 43 °C · 24 h at 27 °C (design)
Weight · battery	—	—	5.5 kg · 200 Wh	3.75 kg · 390 Wh
Charging	needs a freezer	needs a freezer	mains	solar, vehicle or mains
Monitoring	none	none	live tracking	16 sensing points · 4G with offline buffer · per-trip compliance report
WHO PQS prequalification	listed (E004)	varies	listed (E003)	on the E003 route — Emvolio's category — funded by this round

WHO PQS is the prequalification UN agencies and immunisation programmes buy against: E003 covers powered cooling devices, E004 insulated containers
Patent filed · CDSCO (India's medical-device regulator) test licence · no refrigerant at all · five-year cost of ownership about a third below passive, on our model

\$700 device. \$60 a year. 5-year warranty.

Two revenue lines — the device, with a five-year warranty, and a service plan — and, for institutions, the whole thing as a contract.

LINE	PRICE	WHAT'S INCLUDED
Device • 2 L	\$700 today \$500 at scale	Device, charger and five-year warranty. Sold near cost at pilot volumes; at scale, \$500 against a build of about \$200 after enclosures are injection-moulded.
Service plan	\$60 per device per year	4G telemetry, alerts, per-trip compliance reports, maintenance. About 14% of revenue by year five.
Cold-chain-as-a-service	Priced per contract	We carry the payload and own the audit trail; they own the patient. First engagement starting with Roche India, Kolkata: depot to hospital.

\$1,000

Lifetime value per device over five years: \$700 device + 5 × \$60 service.

60%

Gross margin at scale — \$500 price against a ~\$200 build. Near cost today.

B2B

For three years — pharma, diagnostics chains, labs and healthcare institutions, reached through pharma distributors and hospital groups. Consumer comes later, by choice.

Fifty units in service.

All fifty bought.

Forty bought commercially, ten through grant programmes. Beckman Coulter ran diagnostic samples on 7–8 hour routes for six months, reported no excursions and no sample loss — then bought.

96%

of 1,883 monitored hold-hours in 2–8 °C – chamber telemetry from 17 connected units, January–August 2026

0

readings below 0 °C, fleet-wide – nothing freezes

0

excursions reported in six months of 7–8 h sample runs at Beckman Coulter

IN HAND

CDSCO (India's medical-device regulator) test licence · patent filed · EMI/EMC testing at TÜV SÜD to IEC 61326-1 · IP55 dust-and-water ingress, tested in a NABL-accredited lab · founder trained as ISO 13485 internal auditor

NEXT 12 MO

ISO 13485 quality system + CDSCO manufacturing licence (Form MD-5) · WHO PQS E003 submission



Real payload – a blood bag in the 2 L chamber

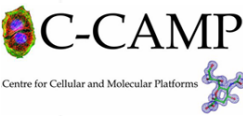
Who buys it. Who backs it.

Beckman Coulter, Roche India and Solumatix are the three commercial anchors (pages 5 and 8). Around them, clinics, hospitals, dairy and research — with units reporting from five Indian states.

CUSTOMERS, PILOTS & DEPLOYMENTS

 Diagnostics • bought after six months	 Cold-chain-as-a-service • Kolkata • starting	 SELCO Foundation Primary-health outreach	 Swami Vivekananda Youth Movement District hospital	 Dairy & veterinary
 Eye care	 Rural health, Rajasthan	 TIGS Tata Institute for Genetics and Society Genetics research	 Screening	 30-unit order • advance received

INCUBATED & FUNDED BY

 SEED fund	 TIDE 2.0	 NIDHI PRAYAS	 Incubation	 Incubation
 Grant	 IQ 2.0 grant	 Leaders in Innovation Fellowship	 DPIIT recognised	 Karnataka • Tamil Nadu • Maharashtra • Gujarat • Uttar Pradesh

We need one percent of this market.

\$1.2 billion addressable; \$0.7 billion in the regions we can reach, the Gulf included. One percent is \$12 million a year by year five, and our plan reaches two-thirds of it — B2B for three years.

\$1.2^B

TAM — about 1 million last-mile cold-chain devices in use worldwide × ~\$1,200 average lifetime value at market prices. Company bottom-up estimate.

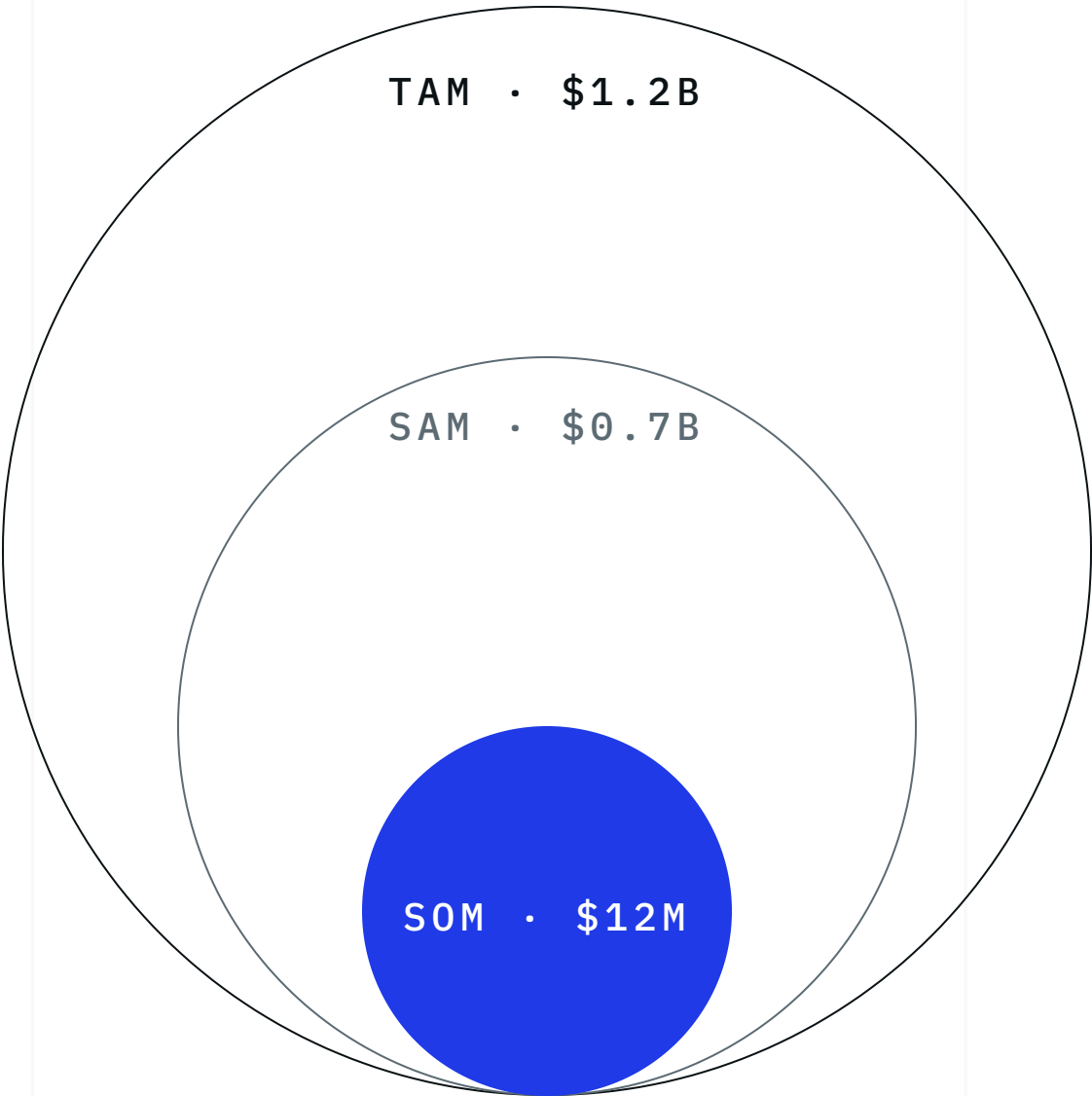
\$0.7^B

SAM — India, the Middle East, Central Asia, Africa and Southeast Asia: the geographies our distributor model can reach.

\$12^M

SOM — 1% of TAM a year by year five, deliberately modest. Our year-five plan of \$7.9M sits inside it.

Context: pharma spends \$21.3B a year on cold-chain logistics (IQVIA, 2025). In Kazakhstan and Uzbekistan about 9 in 10 medicines are imported, so cold-chain products travel a long way to the patient — which is why Central Asia sits inside the SAM. TAM is our separate bottom-up estimate of the last-mile device market.



Certified step by step. Evidence from the field, not the bench.

Medical-device buyers need a path they can audit. This is ours: what is in hand today, what the next twelve months add, and what this round funds.

IN HAND	CDSCO (India's medical-device regulator) test licence · EMI/EMC testing at TÜV SÜD to IEC 61326-1 · IP55 dust-and-water ingress, tested in a NABL-accredited lab · patent filed in India · founder trained as ISO 13485 internal auditor
NEXT 12 MO	ISO 13485 quality system · CDSCO manufacturing licence (Form MD-5) · WHO PQS E003 submission
FUNDED HERE	CE marking for export markets · 30-versus-30 comparative field study against ice-pack carriers under an ethics-committee-approved protocol, data published

96%

of 1,883 monitored hold-hours inside 2–8 °C – chamber telemetry from 17 connected units, January–August 2026

Fleet telemetry



0

readings below 0 °C fleet-wide, and no haemolysis in blood-bag transport at Raktamitra, GIMS Noida

Fleet telemetry · partner report

6^{mo}

of 7–8 h diagnostic-sample runs at Beckman Coulter with no excursions and no sample loss – then a purchase order

Customer report, 2026

From a 50 ml prototype to a deployed platform.

Five years, each version rebuilt from field data — from a bench prototype to ten units with SELCO Foundation to a commercial device. Next: new countries, starting with the Gulf.



2021
Genesis
50 ml solar-powered unit. Proof of concept.



2023
Field trials
10 units deployed with SELCO Foundation.



2024
Version 3.0
Redesigned from field data. CDSCO test licence.



2025
Commercial-ready
Analytics on board. Tested at TÜV SÜD. Patent filed.



2026 →
Scale
2.5 L, 5 L and 10 L. International expansion.

Today the device. Tomorrow the chain.

One cold chain, end to end. Samples reach the lab in a 2 L; doses leave the manufacturer in a 10 L, cross the country in a cold container, then step down — a 5 L on a cycle, a 2 L on a health worker's back — to a child under a tree, with one temperature record the whole way.

THE LONG VIEW

One provider and one temperature record from factory cold room to vaccination session: transport, last-mile devices and site storage, all monitored, all auditable. In this plan: the 2 L today, the 5 L in FY27, the 10 L in FY28. The container and site storage are the ten-to-twenty-year view. The biologics decade needs that chain of custody, and the record that proves it.



Final frame of Phlton's vision film: a health worker hands over a dose at sunrise — "Every dose arrives as it left."

Design, electronics and firmware. All developed in-house.



Suraj J R

CO-FOUNDER

Built the cold engine and next-gen controller

Shruti

FIRMWARE

Embedded and edge-AI control

Ankita Mittal

CEO

ISB • market entry, first institutional deployments

Prerit Mittal

FOUNDER

IIM-A • Royal Academy of Engineering Fellow

Chairman • Harkesh Kumar Mittal – architect of India's national incubation programme (NIDHI); chaired DPIIT's Start-up India seed fund committee

Group portrait composed from the team's individual photographs

\$1.1M at \$8.5M pre-money.

One round to ISO 13485, CE and a 5-litre lab product. Milestones to Series A: a thousand units in service, a \$1M annual revenue run-rate, PQS E003 filed.

35%	Go-to-market — pharma distributors, diagnostics chains and hospital groups in India and the Gulf
25%	Quality and regulatory — ISO 13485, CDSCO MD-5, CE; the first hire is a quality manager
25%	5 L and 10 L manufacturing, the 5 L first, for labs and clinical-trial logistics
15%	IP, WHO PQS E003 submission, working capital

BEYOND CAPITAL

Clinical and laboratory networks for the 5 L; regulatory counsel for CE; introductions to pharma distributors and hospital groups. Manufacturing stays in India; a regional sales-and-service entity follows the first export contract.

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What this round builds – the 5 L platform, FY27